15.3 Double Integrals in Polar Form

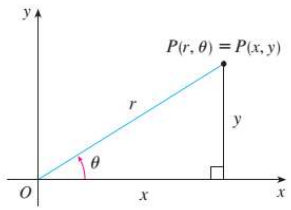


FIGURE 2

Recall from Figure 2 that the polar coordinates (r, θ) of a point are related to the rectangular coordinates (x, y) by the equations

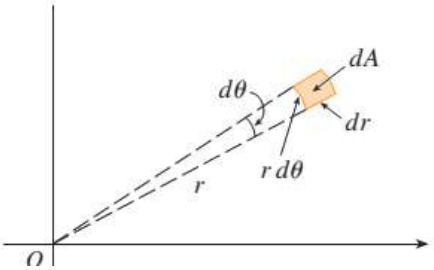
$$r^2 = x^2 + y^2 \qquad x = r\cos\theta \qquad y = r\sin\theta$$

(See Section 10.3.)

The regions in Figure 1 are special cases of a **polar rectangle**

$$R = \{(r, \theta) \mid a \leq r \leq b, \alpha \leq \theta \leq \beta\}$$

The formula in (2) says that we convert from rectangular to polar coordinates in double integral by writing $x = r\cos\theta$ and $y = r\sin\theta$, using the appropriate limits c integration for r and θ, and replacing dA by $r\,dr\,d\theta$. Be careful not to forget the additiona factor r on the right side of Formula 2. A classical method for remembering this is show in Figure 5, where the "infinitesimal" polar rectangle can be thought of as an ordinary rect angle with dimensions $r\,d\theta$ and dr and therefore has "area" $dA = r\,dr\,d\theta$.



Area in Polar Coordinates

The area of a closed and bounded region R in the polar coordinate plane is

$$A = \iint_R r\,dr\,d\theta.$$

2 CHANGE TO POLAR COORDINATES IN A DOUBLE INTEGRAL If f is continuous on a polar rectangle R given by $0 \leq a \leq r \leq b$, $\alpha \leq \theta \leq \beta$, where $0 \leq \beta - \alpha \leq 2\pi$, then

$$\iint_R f(x, y)\,dA = \int_\alpha^\beta \int_a^b f(r\cos\theta, r\sin\theta)\,r\,dr\,d\theta$$

EXAMPLE 1 Evaluate $\iint_R (3x + 4y^2)\,dA$, where R is the region in the upper half-plane bounded by the circles $x^2 + y^2 = 1$ and $x^2 + y^2 = 4$.

SOLUTION The region R can be described as

$$R = \{(x, y) \mid y \geqslant 0,\ 1 \leqslant x^2 + y^2 \leqslant 4\}$$

It is the half-ring shown in Figure 1(b), and in polar coordinates it is given by $1 \leqslant r \leqslant 2$, $0 \leqslant \theta \leqslant \pi$. Therefore, by Formula 2,

$$\begin{aligned}
\iint_R (3x + 4y^2)\,dA &= \int_0^{\pi} \int_1^2 (3r\cos\theta + 4r^2\sin^2\theta)\, r\,dr\,d\theta \\
&= \int_0^{\pi} \int_1^2 (3r^2\cos\theta + 4r^3\sin^2\theta)\,dr\,d\theta \\
&= \int_0^{\pi} \left[r^3\cos\theta + r^4\sin^2\theta\right]_{r=1}^{r=2} d\theta = \int_0^{\pi} (7\cos\theta + 15\sin^2\theta)\,d\theta \\
&= \int_0^{\pi} \left[7\cos\theta + \tfrac{15}{2}(1 - \cos 2\theta)\right] d\theta \\
&= 7\sin\theta + \frac{15\theta}{2} - \frac{15}{4}\sin 2\theta \Big]_0^{\pi} = \frac{15\pi}{2}
\end{aligned}$$

■

EXAMPLE 2 Find the volume of the solid bounded by the plane $z = 0$ and the paraboloid $z = 1 - x^2 - y^2$.

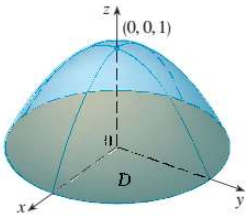


FIGURE 6

SOLUTION If we put $z = 0$ in the equation of the paraboloid, we get $x^2 + y^2 = 1$. This means that the plane intersects the paraboloid in the circle $x^2 + y^2 = 1$, so the solid lies under the paraboloid and above the circular disk D given by $x^2 + y^2 \leqslant 1$ [see Figures 6 and 1(a)]. In polar coordinates D is given by $0 \leqslant r \leqslant 1$, $0 \leqslant \theta \leqslant 2\pi$. Since $1 - x^2 - y^2 = 1 - r^2$, the volume is

$$\begin{aligned}
V &= \iint_D (1 - x^2 - y^2)\,dA = \int_0^{2\pi} \int_0^1 (1 - r^2)\, r\,dr\,d\theta \\
&= \int_0^{2\pi} d\theta \int_0^1 (r - r^3)\,dr = 2\pi\left[\frac{r^2}{2} - \frac{r^4}{4}\right]_0^1 = \frac{\pi}{2}
\end{aligned}$$

If we had used rectangular coordinates instead of polar coordinates, then we would have obtained

$$V = \iint_D (1 - x^2 - y^2)\,dA = \int_{-1}^{1} \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} (1 - x^2 - y^2)\,dy\,dx$$

which is not easy to evaluate because it involves finding $\int (1 - x^2)^{3/2}\,dx$. ■

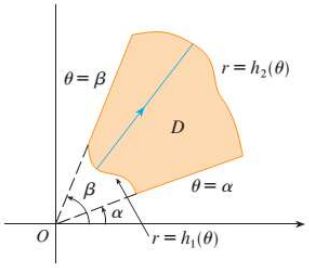


FIGURE 7
$D = \{(r, \theta) \mid \alpha \leq \theta \leq \beta,\ h_1(\theta) \leq r \leq h_2(\theta)\}$

What we have done so far can be extended to the more complicated type of region shown in Figure 7. It's similar to the type II rectangular regions considered in Section 15.3. In fact, by combining Formula 2 in this section with Formula 15.3.5, we obtain the following formula.

3 If f is continuous on a polar region of the form

$$D = \{(r, \theta) \mid \alpha \leq \theta \leq \beta,\ h_1(\theta) \leq r \leq h_2(\theta)\}$$

then

$$\iint_D f(x, y)\, dA = \int_\alpha^\beta \int_{h_1(\theta)}^{h_2(\theta)} f(r\cos\theta, r\sin\theta)\, r\, dr\, d\theta$$

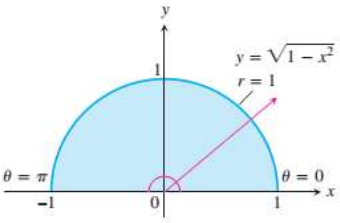


FIGURE 15.26 The semicircular region in Example 4 is the region

$$0 \leq r \leq 1, \qquad 0 \leq \theta \leq \pi.$$

EXAMPLE 4 Evaluating Integrals Using Polar Coordinates

Evaluate

$$\iint_R e^{x^2+y^2}\, dy\, dx,$$

where R is the semicircular region bounded by the x-axis and the curve $y = \sqrt{1 - x^2}$ (Figure 15.26).

Solution In Cartesian coordinates, the integral in question is a nonelementary integral and there is no direct way to integrate $e^{x^2+y^2}$ with respect to either x or y. Yet this integral and others like it are important in mathematics—in statistics, for example—and we need to find a way to evaluate it. Polar coordinates save the day. Substituting $x = r\cos\theta$, $y = r\sin\theta$ and replacing $dy\, dx$ by $r\, dr\, d\theta$ enables us to evaluate the integral as

$$\iint_R e^{x^2+y^2}\, dy\, dx = \int_0^\pi \int_0^1 e^{r^2} r\, dr\, d\theta = \int_0^\pi \left[\frac{1}{2} e^{r^2}\right]_0^1 d\theta$$

$$= \int_0^\pi \frac{1}{2}(e - 1)\, d\theta = \frac{\pi}{2}(e - 1).$$

The r in the $r\, dr\, d\theta$ was just what we needed to integrate e^{r^2}. Without it, we would have been unable to proceed. ■

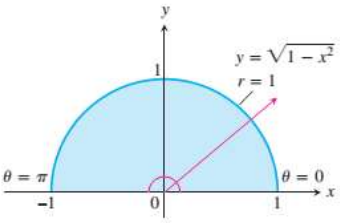


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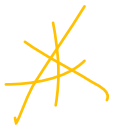
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$$\iint_R e^{x^2+y^2}\, dy\, dx = \int_0^\pi \int_0^1 e^{r^2} r\, dr\, d\theta = \int_0^\pi \left[\frac{1}{2} e^{r^2}\right]_0^1 d\theta$$

$$= \int_0^\pi \frac{1}{2}(e - 1)\, d\theta = \frac{\pi}{2}(e - 1).$$

The r in the $r\, dr\, d\theta$ was just what we needed to integrate e^{r^2}. Without it, we would have been unable to proceed. ■

3.) Find the volume of the solid region $x^2+y^2=x$ that lies inside the solid sphere $x^2+y^2+z^2 \leq$



Solution:

We want to find the volume of the circular cylinder $x^2+y^2=x$ lying inside the unit sphere. First, note that z can be both positive and negative, i.e.,

$$z=\pm\sqrt{1-x^2-y^2}\,.$$

The required volume can be found from the following double integral,

$$V=2\int\int_R \sqrt{1-x^2-y^2}\,dA\,.$$

The factor of 2 appears since the integrand has to be positive for a volume. The region R is the circle $x^2+y^2=x$. Transforming this double integral into polar coordinates we get

$$V=4\int_0^{\pi/2}\int_0^{\cos\theta}\sqrt{1-r^2}\,r\,dr\,d\theta\,.$$

Here we got another factor of 2 using symmetry around x-axis. Performing the radial integration we find

$$V=\frac{4}{3}\int_0^{\pi/2}(1-\sin^3\theta)\,d\theta=\frac{4}{3}\int_0^{\pi/2}(1-\sin\theta+\sin\theta\cos^2\theta)\,d\theta=\frac{6\pi-8}{9}\,.$$

15.4 Triple Integrals in Rectangular Coordinates

$$\lim_{n\to\infty}\sum_{k=1}^{n}\Delta V_k=\iiint_D dV.$$

DEFINITION **Volume**

The **volume** of a closed, bounded region D in space is

$$V=\iiint_D dV.$$

4 FUBINI'S THEOREM FOR TRIPLE INTEGRALS If f is continuous on the rectangular box $B = [a, b] \times [c, d] \times [r, s]$, then

$$\iiint_B f(x, y, z)\, dV = \int_r^s \int_c^d \int_a^b f(x, y, z)\, dx\, dy\, dz$$

Finding Limits of Integration

We evaluate a triple integral by applying a three-dimensional version of Fubini's Theorem (Section 15.1) to evaluate it by three repeated single integrations. As with double integrals, there is a geometric procedure for finding the limits of integration for these single integrals.

To evaluate

$$\iiint_D F(x, y, z)\, dV$$

over a region D, integrate first with respect to z, then with respect to y, finally with x.

1. *Sketch:* Sketch the region D along with its "shadow" R (vertical projection) in the xy-plane. Label the upper and lower bounding surfaces of D and the upper and lower bounding curves of R.

2. *Find the z-limits of integration:* Draw a line M passing through a typical point (x, y) in R parallel to the z-axis. As z increases, M enters D at $z = f_1(x, y)$ and leaves at $z = f_2(x, y)$. These are the z-limits of integration.

4. *Find the x-limits of integration:* Choose x-limits that include all lines through R parallel to the y-axis ($x = a$ and $x = b$ in the preceding figure). These are the x-limits of integration. The integral is

$$\int_{x=a}^{x=b} \int_{y=g_1(x)}^{y=g_2(x)} \int_{z=f_1(x,y)}^{z=f_2(x,y)} F(x, y, z)\, dz\, dy\, dx.$$

Follow similar procedures if you change the order of integration. The "shadow" of region D lies in the plane of the last two variables with respect to which the iterated integration takes place.

V EXAMPLE 1 Evaluate the triple integral $\iiint_B xyz^2\,dV$, where B is the rectangular box given by

$$B = \{(x, y, z) \mid 0 \leqslant x \leqslant 1,\ -1 \leqslant y \leqslant 2,\ 0 \leqslant z \leqslant 3\}$$

SOLUTION We could use any of the six possible orders of integration. If we choose to integrate with respect to x, then y, and then z, we obtain

$$\begin{aligned}\iiint_B xyz^2\,dV &= \int_0^3 \int_{-1}^2 \int_0^1 xyz^2\,dx\,dy\,dz = \int_0^3 \int_{-1}^2 \left[\frac{x^2yz^2}{2}\right]_{x=0}^{x=1} dy\,dz \\ &= \int_0^3 \int_{-1}^2 \frac{yz^2}{2}\,dy\,dz = \int_0^3 \left[\frac{y^2z^2}{4}\right]_{y=-1}^{y=2} dz \\ &= \int_0^3 \frac{3z^2}{4}\,dz = \frac{z^3}{4}\Big]_0^3 = \frac{27}{4}\end{aligned}$$

EXAMPLE 1 Finding a Volume

Find the volume of the region D enclosed by the surfaces $z = x^2 + 3y^2$ and $z = 8 - x^2 - y^2$.

Solution The volume is

$$V = \iiint_D dz\,dy\,dx,$$

the integral of $F(x, y, z) = 1$ over D. To find the limits of integration for evaluating the integral, we first sketch the region. The surfaces (Figure 15.28) intersect on the elliptical cylinder $x^2 + 3y^2 = 8 - x^2 - y^2$ or $x^2 + 2y^2 = 4$, $z > 0$. The boundary of the region R, the projection of D onto the xy-plane, is an ellipse with the same equation: $x^2 + 2y^2 = 4$. The "upper" boundary of R is the curve $y = \sqrt{(4 - x^2)/2}$. The lower boundary is the curve $y = -\sqrt{(4 - x^2)/2}$.

Finally we find the x-limits of integration. As L sweeps across R, the value of x varies from $x = -2$ at $(-2, 0, 0)$ to $x = 2$ at $(2, 0, 0)$. The volume of D is

$$
\begin{aligned}
V &= \iiint_D dz\,dy\,dx \\
&= \int_{-2}^{2} \int_{-\sqrt{(4-x^2)/2}}^{\sqrt{(4-x^2)/2}} \int_{x^2+3y^2}^{8-x^2-y^2} dz\,dy\,dx \\
&= \int_{-2}^{2} \int_{-\sqrt{(4-x^2)/2}}^{\sqrt{(4-x^2)/2}} (8 - 2x^2 - 4y^2)\,dy\,dx \\
&= \int_{-2}^{2} \left[(8 - 2x^2)y - \frac{4}{3}y^3 \right]_{y=-\sqrt{(4-x^2)/2}}^{y=\sqrt{(4-x^2)/2}} dx \\
&= \int_{-2}^{2} \left(2(8 - 2x^2)\sqrt{\frac{4-x^2}{2}} - \frac{8}{3}\left(\frac{4-x^2}{2}\right)^{3/2} \right) dx \\
&= \int_{-2}^{2} \left[8\left(\frac{4-x^2}{2}\right)^{3/2} - \frac{8}{3}\left(\frac{4-x^2}{2}\right)^{3/2} \right] dx = \frac{4\sqrt{2}}{3} \int_{-2}^{2} (4 - x^2)^{3/2}\,dx \\
&= 8\pi\sqrt{2}. \qquad \text{After integration with the substitution } x = 2\sin u\,.
\end{aligned}
$$

■

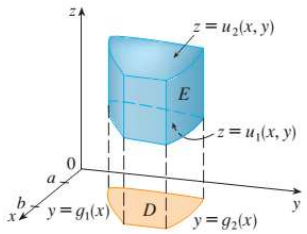


FIGURE 3
A type 1 solid region where the projection D is a type I plane region

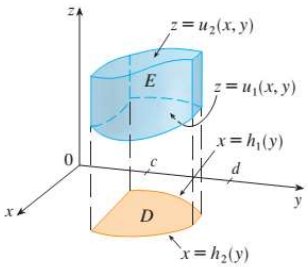
The meaning of the inner integral on the right side of Equation 6 is that x and y are held fixed, and therefore $u_1(x, y)$ and $u_2(x, y)$ are regarded as constants, while $f(x, y, z)$ is integrated with respect to z.

In particular, if the projection D of E onto the xy-plane is a type I plane region (as in Figure 3), then

$$E = \{(x, y, z) \mid a \leq x \leq b,\ g_1(x) \leq y \leq g_2(x),\ u_1(x, y) \leq z \leq u_2(x, y)\}$$

and Equation 6 becomes

$$\boxed{7} \qquad \iiint_E f(x, y, z)\,dV = \int_a^b \int_{g_1(x)}^{g_2(x)} \int_{u_1(x,y)}^{u_2(x,y)} f(x, y, z)\,dz\,dy\,dx$$



If, on the other hand, D is a type II plane region (as in Figure 4), then

$$E = \{(x, y, z) \mid c \leq y \leq d,\ h_1(y) \leq x \leq h_2(y),\ u_1(x, y) \leq z \leq u_2(x, y)\}$$

and Equation 6 becomes

$$\boxed{8} \qquad \iiint_E f(x, y, z)\,dV = \int_c^d \int_{h_1(y)}^{h_2(y)} \int_{u_1(x,y)}^{u_2(x,y)} f(x, y, z)\,dz\,dx\,dy$$

EXAMPLE 2 Evaluate $\iiint_E z\,dV$, where E is the solid tetrahedron bounded by the four planes $x = 0$, $y = 0$, $z = 0$, and $x + y + z = 1$.

SOLUTION When we set up a triple integral it's wise to draw *two* diagrams: one of the solid region E (see Figure 5) and one of its projection D on the xy-plane (see Figure 6). The lower boundary of the tetrahedron is the plane $z = 0$ and the upper

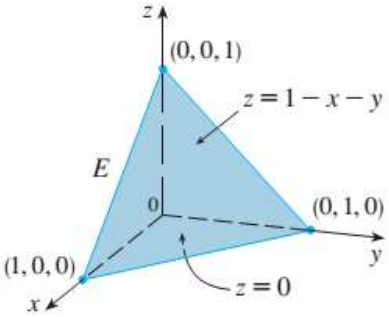


FIGURE 5

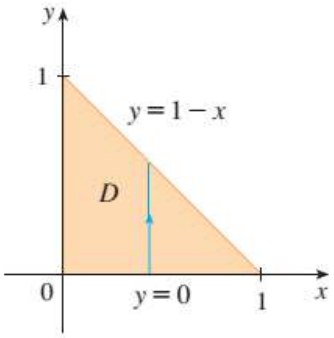


FIGURE 6

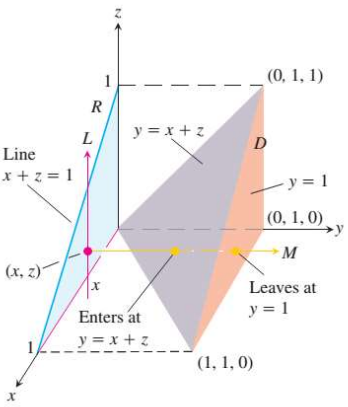
$$\boxed{9} \qquad E = \{(x, y, z) \mid 0 \leqslant x \leqslant 1,\ 0 \leqslant y \leqslant 1 - x,\ 0 \leqslant z \leqslant 1 - x - y\}$$

This description of E as a type 1 region enables us to evaluate the integral as follows:

$$\iiint_E z\,dV = \int_0^1 \int_0^{1-x} \int_0^{1-x-y} z\,dz\,dy\,dx = \int_0^1 \int_0^{1-x} \left[\frac{z^2}{2}\right]_{z=0}^{z=1-x-y} dy\,dx$$

$$= \tfrac{1}{2}\int_0^1 \int_0^{1-x} (1 - x - y)^2\,dy\,dx = \tfrac{1}{2}\int_0^1 \left[-\frac{(1 - x - y)^3}{3}\right]_{y=0}^{y=1-x} dx$$

$$= \tfrac{1}{6}\int_0^1 (1 - x)^3\,dx = \frac{1}{6}\left[-\frac{(1 - x)^4}{4}\right]_0^1 = \frac{1}{24}$$



EXAMPLE 2 Finding the Limits of Integration in the Order *dy dz dx*

Set up the limits of integration for evaluating the triple integral of a function $F(x, y, z)$ over the tetrahedron D with vertices (0, 0, 0), (1, 1, 0), (0, 1, 0), and (0, 1, 1).

Solution We sketch D along with its "shadow" R in the xz-plane (Figure 15.29). The upper (right-hand) bounding surface of D lies in the plane $y = 1$. The lower (left-hand) bounding surface lies in the plane $y = x + z$. The upper boundary of R is the line $z = 1 - x$. The lower boundary is the line $z = 0$.

First we find the y-limits of integration. The line through a typical point (x, z) in R parallel to the y-axis enters D at $y = x + z$ and leaves at $y = 1$.

Next we find the z-limits of integration. The line L through (x, z) parallel to the z-axis enters R at $z = 0$ and leaves at $z = 1 - x$.

Finally we find the x-limits of integration. As L sweeps across R, the value of x varies from $x = 0$ to $x = 1$. The integral is

$$\int_0^1 \int_0^{1-x} \int_{x+z}^1 F(x, y, z)\,dy\,dz\,dx.$$

EXAMPLE 3 Revisiting Example 2 Using the Order *dz dy dx*

$$\int_0^1 \int_x^1 \int_0^{y-x} F(x, y, z)\, dz\, dy\, dx.$$

For example, if $F(x, y, z) = 1$, we would find the volume of the tetrahedron to be

$$\begin{aligned} V &= \int_0^1 \int_x^1 \int_0^{y-x} dz\, dy\, dx \\ &= \int_0^1 \int_x^1 (y - x)\, dy\, dx \\ &= \int_0^1 \left[\frac{1}{2}y^2 - xy\right]_{y=x}^{y=1} dx \\ &= \int_0^1 \left(\frac{1}{2} - x + \frac{1}{2}x^2\right) dx \\ &= \left[\frac{1}{2}x - \frac{1}{2}x^2 + \frac{1}{6}x^3\right]_0^1 \\ &= \frac{1}{6}. \end{aligned}$$

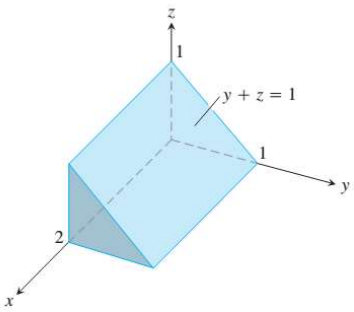


FIGURE 15.30 Example 4 gives six different iterated triple integrals for the volume of this prism.

EXAMPLE 4 Using Different Orders of Integration

Each of the following integrals gives the volume of the solid shown in Figure 15.30.

(a) $\int_0^1 \int_0^{1-z} \int_0^2 dx\, dy\, dz$ **(b)** $\int_0^1 \int_0^{1-y} \int_0^2 dx\, dz\, dy$

(c) $\int_0^1 \int_0^2 \int_0^{1-z} dy\, dx\, dz$ **(d)** $\int_0^2 \int_0^1 \int_0^{1-z} dy\, dz\, dx$

(e) $\int_0^1 \int_0^2 \int_0^{1-y} dz\, dx\, dy$ **(f)** $\int_0^2 \int_0^1 \int_0^{1-y} dz\, dy\, dx$

We work out the integrals in parts (b) and (c):

$$V = \int_0^1 \int_0^{1-y} \int_0^2 dx\, dz\, dy \qquad \text{Integral in part (b)}$$

A solid region E is of **type 2** if it is of the form

$$E = \{(x, y, z) \mid (y, z) \in D,\ u_1(y, z) \leq x \leq u_2(y, z)\}$$

where, this time, D is the projection of E onto the yz-plane (see Figure 7). The back surface is $x = u_1(y, z)$, the front surface is $x = u_2(y, z)$, and we have

10
$$\iiint_E f(x, y, z)\, dV = \iint_D \left[\int_{u_1(y,z)}^{u_2(y,z)} f(x, y, z)\, dx \right] dA$$

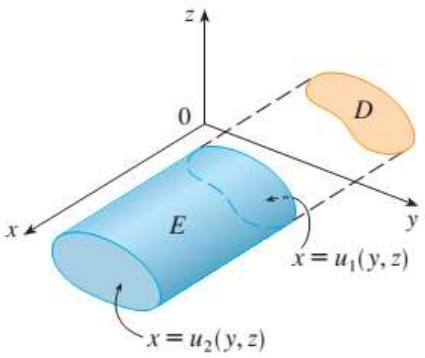


FIGURE 7 A type 2 region

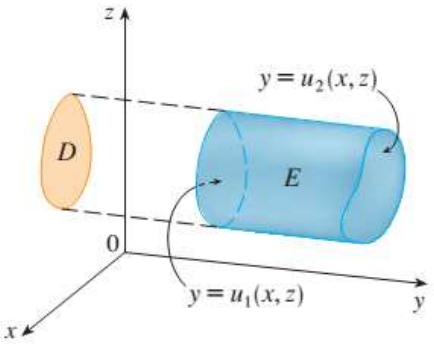


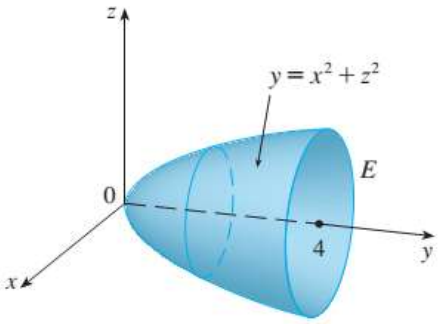
FIGURE 8 A type 3 region

Finally, a **type 3** region is of the form

$$E = \{(x, y, z) \mid (x, z) \in D,\ u_1(x, z) \leq y \leq u_2(x, z)\}$$

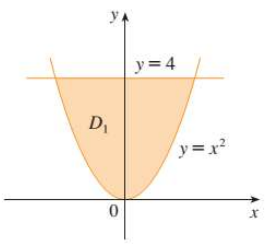
EXAMPLE 3 Evaluate $\iiint_E \sqrt{x^2 + z^2}\, dV$, where E is the region bounded by the paraboloid $y = x^2 + z^2$ and the plane $y = 4$.

SOLUTION The solid E is shown in Figure 9. If we regard it as a type 1 region, then we need to consider its projection D_1 onto the xy-plane, which is the parabolic region in Figure 10. (The trace of $y = x^2 + z^2$ in the plane $z = 0$ is the parabola $y = x^2$.)



9

on



From $y = x^2 + z^2$ we obtain $z = \pm\sqrt{y - x^2}$, so the lower boundary surface of E is $z = -\sqrt{y - x^2}$ and the upper surface is $z = \sqrt{y - x^2}$. Therefore the description of E as a type 1 region is

$$E = \{(x, y, z) \mid -2 \leq x \leq 2,\ x^2 \leq y \leq 4,\ -\sqrt{y - x^2} \leq z \leq \sqrt{y - x^2}\}$$

and so we obtain

$$\iiint_E \sqrt{x^2 + z^2}\, dV = \int_{-2}^{2} \int_{x^2}^{4} \int_{-\sqrt{y-x^2}}^{\sqrt{y-x^2}} \sqrt{x^2 + z^2}\, dz\, dy\, dx$$

FIGURE 10
Projection on xy-plane

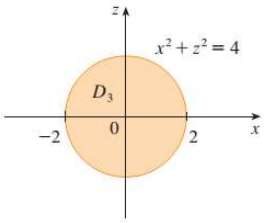


FIGURE 11
Projection on xz-plane

The most difficult step in evaluating a triple integral is setting up an expression for the region of integration (such as Equation 9 in Example 2). Remember that the limits of integration in the inner integral contain at most two variables, the limits of integration in the middle integral contain at most one variable, and the limits of integration in the outer integral must be constants.

Although this expression is correct, it is extremely difficult to evaluate. So let's instead consider E as a type 3 region. As such, its projection D_3 onto the xz-plane is the disk $x^2 + z^2 \leqslant 4$ shown in Figure 11.

Then the left boundary of E is the paraboloid $y = x^2 + z^2$ and the right boundary is the plane $y = 4$, so taking $u_1(x, z) = x^2 + z^2$ and $u_2(x, z) = 4$ in Equation 11, we have

$$\iiint_E \sqrt{x^2 + z^2}\, dV = \iint_{D_3} \left[\int_{x^2+z^2}^{4} \sqrt{x^2 + z^2}\, dy \right] dA = \iint_{D_3} (4 - x^2 - z^2)\sqrt{x^2 + z^2}\, dA$$

Although this integral could be written as

$$\int_{-2}^{2} \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} (4 - x^2 - z^2)\sqrt{x^2 + z^2}\, dz\, dx$$

it's easier to convert to polar coordinates in the xz-plane: $x = r\cos\theta$, $z = r\sin\theta$. This gives

$$\begin{aligned} \iiint_E \sqrt{x^2 + z^2}\, dV &= \iint_{D_3} (4 - x^2 - z^2)\sqrt{x^2 + z^2}\, dA \\ &= \int_0^{2\pi} \int_0^2 (4 - r^2) r\, r\, dr\, d\theta = \int_0^{2\pi} d\theta \int_0^2 (4r^2 - r^4)\, dr \\ &= 2\pi \left[\frac{4r^3}{3} - \frac{r^5}{5} \right]_0^2 = \frac{128\pi}{15} \end{aligned}$$

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$$V(E) = \iiint_E dV$$

For example, you can see this in the case of a type 1 region by putting $f(x, y, z) = 1$ in Formula 6:

$$\iiint_E 1\, dV = \iint_D \left[\int_{u_1(x,y)}^{u_2(x,y)} dz \right] dA = \iint_D [u_2(x, y) - u_1(x, y)]\, dA$$

and from Section 15.3 we know this represents the volume that lies between the surfaces $z = u_1(x, y)$ and $z = u_2(x, y)$.

EXAMPLE 4 Use a triple integral to find the volume of the tetrahedron T bounded by the planes $x + 2y + z = 2$, $x = 2y$, $x = 0$, and $z = 0$.

SOLUTION The tetrahedron T and its projection D on the xy-plane are shown in Figures 12 and 13. The lower boundary of T is the plane $z = 0$ and the upper boundary is the plane $x + 2y + z = 2$, that is, $z = 2 - x - 2y$.

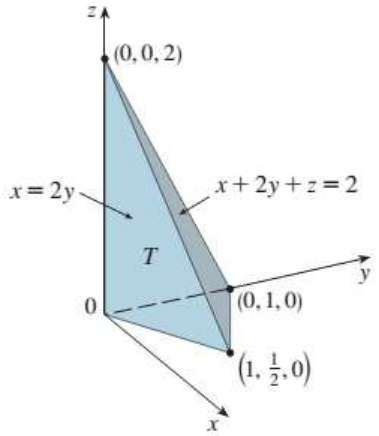


FIGURE 12

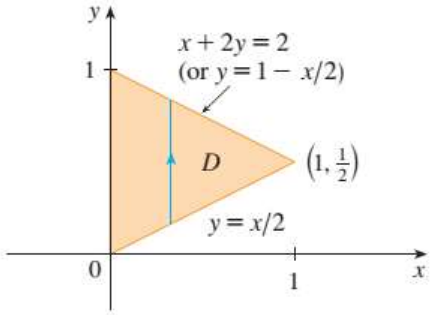


FIGURE 13

Therefore we have

$$V(T) = \iiint_T dV = \int_0^1 \int_{x/2}^{1-x/2} \int_0^{2-x-2y} dz\, dy\, dx$$

$$= \int_0^1 \int_{x/2}^{1-x/2} (2 - x - 2y)\, dy\, dx = \tfrac{1}{3}$$

5.) b) Express the volume of the region R bounded by the parabolic cylinder $x = y^2$ and the planes $z = 0$ and $x + z = 1$ as a triple integral in the orders $dzdxdy$ and $dydzdx$. (Do not evaluate the integrals.)

Solution:

The projection of the region on the xy-plane is $x = y^2$, and $0 \le x \le 1$. So, the volume is

$$V = \int_{-1}^{1} \int_{y^2}^{1} \int_{0}^{1-x} dzdxdy$$

The projection of the region on the zx-plane is $x + z = 1$ and $0 \le x \le 1$ which implies

$$V = \int_{0}^{1} \int_{0}^{1-x} \int_{-\sqrt{x}}^{\sqrt{x}} dydzdx$$

15.6 Triple Integrals in Cylindrical and Spherical Coordinates

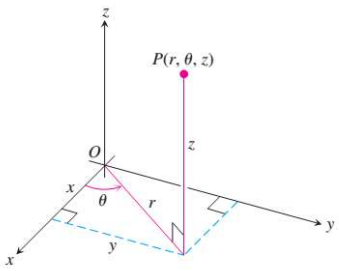


FIGURE 15.36 The cylindrical coordinates of a point in space are r, θ, and z.

Integration in Cylindrical Coordinates

We obtain cylindrical coordinates for space by combining polar coordinates in the xy-plane with the usual z-axis. This assigns to every point in space one or more coordinate triples of the form (r, θ, z), as shown in Figure 15.36.

DEFINITION Cylindrical Coordinates

Cylindrical coordinates represent a point P in space by ordered triples (r, θ, z) in which

1. r and θ are polar coordinates for the vertical projection of P on the xy-plane
2. z is the rectangular vertical coordinate.

The values of x, y, r, and θ in rectangular and cylindrical coordinates are related by the usual equations.

Equations Relating Rectangular (x, y, z) and Cylindrical (r, θ, z) Coordinates

$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad z = z,$$
$$r^2 = x^2 + y^2, \qquad \tan\theta = y/x$$

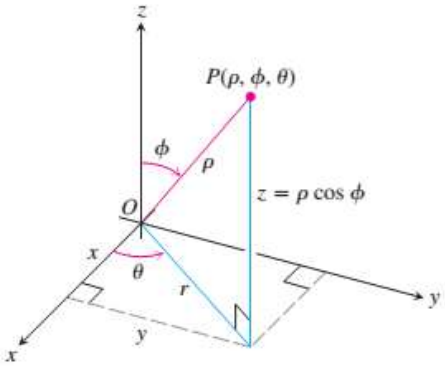


FIGURE 15.41 The spherical coordinates ρ, ϕ, and θ and their relation to x, y, z, and r.

Equations Relating Spherical Coordinates to Cartesian and Cylindrical Coordinates

$$r = \rho\sin\phi, \qquad x = r\cos\theta = \rho\sin\phi\cos\theta,$$
$$z = \rho\cos\phi, \qquad y = r\sin\theta = \rho\sin\phi\sin\theta, \tag{1}$$
$$\rho = \sqrt{x^2 + y^2 + z^2} = \sqrt{r^2 + z^2}.$$

15.8 TRIPLE INTEGRALS IN SPHERICAL COORDINATES

SPHERICAL COORDINATES

The **spherical coordinates** (ρ, θ, ϕ) of a point P in space are shown in Figure 1, where $\rho = |OP|$ is the distance from the origin to P, θ is the same angle as in cylindrical coordinates, and ϕ is the angle between the positive z-axis and the line segment OP. Note that

$$\rho \geqslant 0 \qquad 0 \leqslant \phi \leqslant \pi$$

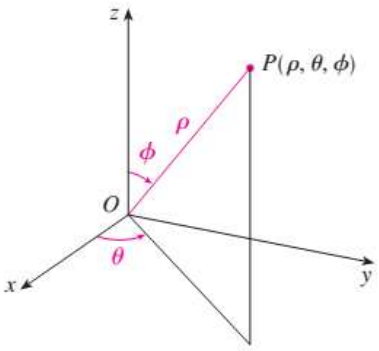


Figure 4).

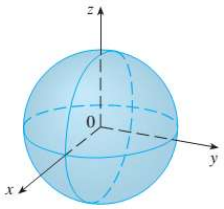


FIGURE 2 $\rho = c$, a sphere

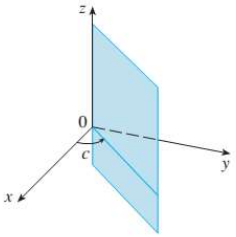


FIGURE 3 $\theta = c$, a half-plane

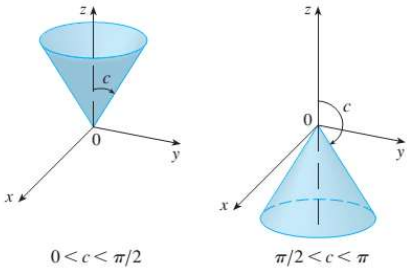


FIGURE 4 $\phi = c$, a half-cone

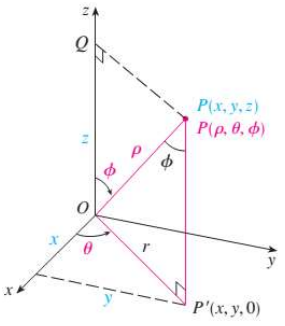


FIGURE 5

The relationship between rectangular and spherical coordinates can be seen from Figure 5. From triangles OPQ and OPP' we have

$$z = \rho \cos \phi \qquad r = \rho \sin \phi$$

But $x = r \cos \theta$ and $y = r \sin \theta$, so to convert from spherical to rectangular coordinates, we use the equations

1 $$x = \rho \sin \phi \cos \theta \qquad y = \rho \sin \phi \sin \theta \qquad z = \rho \cos \phi$$

Also, the distance formula shows that

2 $$\rho^2 = x^2 + y^2 + z^2$$

We use this equation in converting from rectangular to spherical coordinates.

EXAMPLE 1 The point $(2, \pi/4, \pi/3)$ is given in spherical coordinates. Plot the point and find its rectangular coordinates.

SOLUTION We plot the point in Figure 6. From Equations 1 we have

$$x = \rho \sin\phi \cos\theta = 2 \sin\frac{\pi}{3} \cos\frac{\pi}{4} = 2\left(\frac{\sqrt{3}}{2}\right)\left(\frac{1}{\sqrt{2}}\right) = \sqrt{\frac{3}{2}}$$

$$y = \rho \sin\phi \sin\theta = 2 \sin\frac{\pi}{3} \sin\frac{\pi}{4} = 2\left(\frac{\sqrt{3}}{2}\right)\left(\frac{1}{\sqrt{2}}\right) = \sqrt{\frac{3}{2}}$$

$$z = \rho \cos\phi = 2 \cos\frac{\pi}{3} = 2\left(\tfrac{1}{2}\right) = 1$$

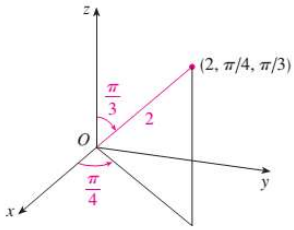


FIGURE 6

Thus the point $(2, \pi/4, \pi/3)$ is $\left(\sqrt{3/2}, \sqrt{3/2}, 1\right)$ in rectangular coordinates. ■

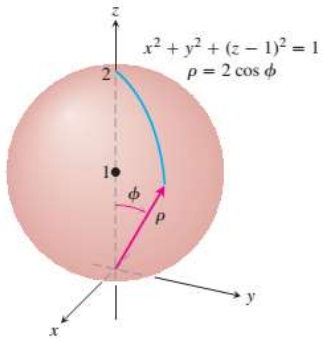


FIGURE 15.43 The sphere in Example 3.

EXAMPLE 3 Converting Cartesian to Spherical

Find a spherical coordinate equation for the sphere $x^2 + y^2 + (z - 1)^2 = 1$.

Solution We use Equations (1) to substitute for x, y, and z:

$$x^2 + y^2 + (z - 1)^2 = 1$$

$$\rho^2 \sin^2\phi \cos^2\theta + \rho^2 \sin^2\phi \sin^2\theta + (\rho\cos\phi - 1)^2 = 1 \qquad \text{Equations (1)}$$

$$\rho^2 \sin^2\phi \underbrace{(\cos^2\theta + \sin^2\theta)}_{1} + \rho^2\cos^2\phi - 2\rho\cos\phi + 1 = 1$$

$$\rho^2 \underbrace{(\sin^2\phi + \cos^2\phi)}_{1} = 2\rho\cos\phi$$

$$\rho^2 = 2\rho\cos\phi$$

$$\rho = 2\cos\phi\,.$$

How to Integrate in Cylindrical Coordinates

To evaluate

$$\iiint f(r, \theta, z)\, dV$$

EVALUATING TRIPLE INTEGRALS WITH CYLINDRICAL COORDINATES

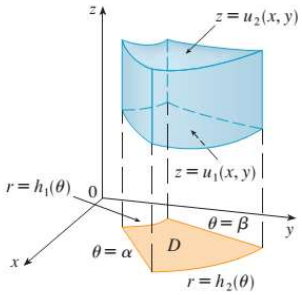


FIGURE 6

Suppose that E is a type 1 region whose projection D on the xy-plane is conveniently described in polar coordinates (see Figure 6). In particular, suppose that f is continuous and

$$E = \{(x, y, z) \mid (x, y) \in D,\ u_1(x, y) \leq z \leq u_2(x, y)\}$$

where D is given in polar coordinates by

$$D = \{(r, \theta) \mid \alpha \leq \theta \leq \beta,\ h_1(\theta) \leq r \leq h_2(\theta)\}$$

We know from Equation 15.6.6 that

$$\boxed{3} \qquad \iiint_E f(x, y, z)\, dV = \iint_D \left[\int_{u_1(x, y)}^{u_2(x, y)} f(x, y, z)\, dz \right] dA$$

But we also know how to evaluate double integrals in polar coordinates. In fact, combining Equation 3 with Equation 15.4.3, we obtain

$$\boxed{4} \qquad \iiint_E f(x, y, z)\, dV = \int_\alpha^\beta \int_{h_1(\theta)}^{h_2(\theta)} \int_{u_1(r\cos\theta,\, r\sin\theta)}^{u_2(r\cos\theta,\, r\sin\theta)} f(r\cos\theta, r\sin\theta, z)\, r\, dz\, dr\, d\theta$$

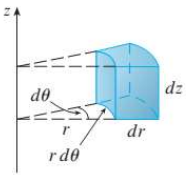


FIGURE 7
Volume element in cylindrical coordinates: $dV = r\, dz\, dr\, d\theta$

Formula 4 is the **formula for triple integration in cylindrical coordinates**. It says that we convert a triple integral from rectangular to cylindrical coordinates by writing $x = r\cos\theta$, $y = r\sin\theta$, leaving z as it is, using the appropriate limits of integration for z, r, and θ, and replacing dV by $r\, dz\, dr\, d\theta$. (Figure 7 shows how to remember this.) It is worthwhile to use this formula when E is a solid region easily described in cylindrical coordinates, and especially when the function $f(x, y, z)$ involves the expression $x^2 + y^2$.

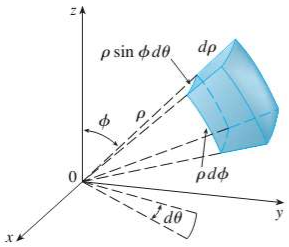


FIGURE 8
Volume element in spherical coordinates: $dV = \rho^2 \sin\phi\, d\rho\, d\theta\, d\phi$

$$\boxed{3} \quad \iiint_E f(x, y, z)\, dV = \int_c^d \int_\alpha^\beta \int_a^b f(\rho\sin\phi\cos\theta, \rho\sin\phi\sin\theta, \rho\cos\phi)\, \rho^2 \sin\phi\, d\rho\, d\theta\, d\phi$$

where E is a spherical wedge given by

$$E = \{(\rho, \theta, \phi) \mid a \leq \rho \leq b,\ \alpha \leq \theta \leq \beta,\ c \leq \phi \leq d\}$$

Formula 3 says that we convert a triple integral from rectangular coordinates to spherical coordinates by writing

$$x = \rho\sin\phi\cos\theta \qquad y = \rho\sin\phi\sin\theta \qquad z = \rho\cos\phi$$

using the appropriate limits of integration, and replacing dV by $\rho^2 \sin\phi\, d\rho\, d\theta\, d\phi$. This is illustrated in Figure 8.

This formula can be extended to include more general spherical regions such as

$$E = \{(\rho, \theta, \phi) \mid \alpha \leq \theta \leq \beta,\ c \leq \phi \leq d,\ g_1(\theta, \phi) \leq \rho \leq g_2(\theta, \phi)\}$$

EXAMPLE 4 Evaluate $\int_{-2}^{2}\int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}}\int_{\sqrt{x^2+y^2}}^{2}(x^2+y^2)\,dz\,dy\,dx$.

SOLUTION This iterated integral is a triple integral over the solid region

$$E=\left\{(x,y,z)\mid -2\leqslant x\leqslant 2,\ -\sqrt{4-x^2}\leqslant y\leqslant\sqrt{4-x^2},\ \sqrt{x^2+y^2}\leqslant z\leqslant 2\right\}$$

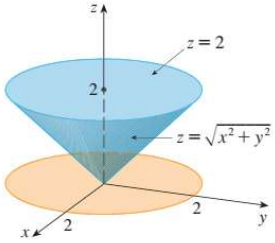


FIGURE 9

and the projection of E onto the xy-plane is the disk $x^2+y^2\leqslant 4$. The lower surface of E is the cone $z=\sqrt{x^2+y^2}$ and its upper surface is the plane $z=2$. (See Figure 9.) This region has a much simpler description in cylindrical coordinates:

$$E=\left\{(r,\theta,z)\mid 0\leqslant\theta\leqslant 2\pi,\ 0\leqslant r\leqslant 2,\ r\leqslant z\leqslant 2\right\}$$

Therefore, we have

$$\begin{aligned}\int_{-2}^{2}\int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}}\int_{\sqrt{x^2+y^2}}^{2}(x^2+y^2)\,dz\,dy\,dx &= \iiint_E (x^2+y^2)\,dV\\ &=\int_0^{2\pi}\int_0^2\int_r^2 r^2\,r\,dz\,dr\,d\theta\\ &=\int_0^{2\pi}d\theta\int_0^2 r^3(2-r)\,dr\\ &=2\pi\left[\tfrac{1}{2}r^4-\tfrac{1}{5}r^5\right]_0^2=\tfrac{16}{5}\pi\end{aligned}$$

■

EXAMPLE 3 Evaluate $\iiint_B e^{(x^2+y^2+z^2)^{3/2}}\,dV$, where B is the unit ball:

$$B=\left\{(x,y,z)\mid x^2+y^2+z^2\leqslant 1\right\}$$

SOLUTION Since the boundary of B is a sphere, we use spherical coordinates:

$$B=\left\{(\rho,\theta,\phi)\mid 0\leqslant\rho\leqslant 1,\ 0\leqslant\theta\leqslant 2\pi,\ 0\leqslant\phi\leqslant\pi\right\}$$

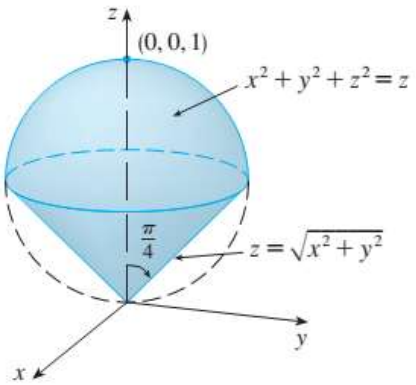
In addition, spherical coordinates are appropriate because

$$x^2+y^2+z^2=\rho^2$$

Thus (3) gives

$$\begin{aligned}\iiint_B e^{(x^2+y^2+z^2)^{3/2}}\,dV &= \int_0^{\pi}\int_0^{2\pi}\int_0^1 e^{(\rho^2)^{3/2}}\rho^2\sin\phi\,d\rho\,d\theta\,d\phi\\ &=\int_0^{\pi}\sin\phi\,d\phi\int_0^{2\pi}d\theta\int_0^1\rho^2e^{\rho^3}d\rho\\ &=\left[-\cos\phi\right]_0^{\pi}(2\pi)\left[\tfrac{1}{3}e^{\rho^3}\right]_0^1=\tfrac{4}{3}\pi(e-1)\end{aligned}$$

V EXAMPLE 4 Use spherical coordinates to find the volume of the solid that lies above the cone $z = \sqrt{x^2 + y^2}$ and below the sphere $x^2 + y^2 + z^2 = z$. (See Figure 9.)



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■ Figure 10 gives another look (this time drawn by Maple) at the solid of Example 4.

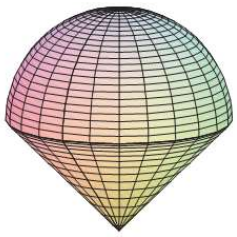


FIGURE 10

SOLUTION Notice that the sphere passes through the origin and has center $\left(0, 0, \frac{1}{2}\right)$. We write the equation of the sphere in spherical coordinates as

$$\rho^2 = \rho \cos \phi \qquad \text{or} \qquad \rho = \cos \phi$$

The equation of the cone can be written as

$$\rho \cos \phi = \sqrt{\rho^2 \sin^2\phi \cos^2\theta + \rho^2 \sin^2\phi \sin^2\theta} = \rho \sin \phi$$

This gives $\sin \phi = \cos \phi$, or $\phi = \pi/4$. Therefore the description of the solid E in spherical coordinates is

$$E = \{(\rho, \theta, \phi) \mid 0 \leq \theta \leq 2\pi,\ 0 \leq \phi \leq \pi/4,\ 0 \leq \rho \leq \cos \phi\}$$

Figure 11 shows how E is swept out if we integrate first with respect to ρ, then ϕ, and then θ. The volume of E is

$$\begin{aligned} V(E) &= \iiint_E dV = \int_0^{2\pi} \int_0^{\pi/4} \int_0^{\cos\phi} \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta \\ &= \int_0^{2\pi} d\theta \int_0^{\pi/4} \sin\phi \left[\frac{\rho^3}{3} \right]_{\rho=0}^{\rho=\cos\phi} d\phi \\ &= \frac{2\pi}{3} \int_0^{\pi/4} \sin\phi \cos^3\phi \, d\phi = \frac{2\pi}{3} \left[-\frac{\cos^4\phi}{4} \right]_0^{\pi/4} = \frac{\pi}{8} \end{aligned}$$

TEC Visual 15.8 shows an animation of Figure 11.

15.9 CHANGE OF VARIABLES IN MULTIPLE INTEGRALS

More generally, we consider a change of variables that is given by a **transformation** T from the uv-plane to the xy-plane:

$$T(u, v) = (x, y)$$

where x and y are related to u and v by the equations

3 $$x = g(u, v) \qquad y = h(u, v)$$

or, as we sometimes write,

$$x = x(u, v) \qquad y = y(u, v)$$

A transformation T is really just a function whose domain and range are both subsets of $\mathbb{R}^2$. If $T(u_1, v_1) = (x_1, y_1)$, then the point (x_1, y_1) is called the **image** of the point (u_1, v_1). If no two points have the same image, T is called **one-to-one**. Figure 1 shows the effect of a transformation T on a region S in the uv-plane. T transforms S into a region R in the xy-plane called the **image of S**, consisting of the images of all points in S.

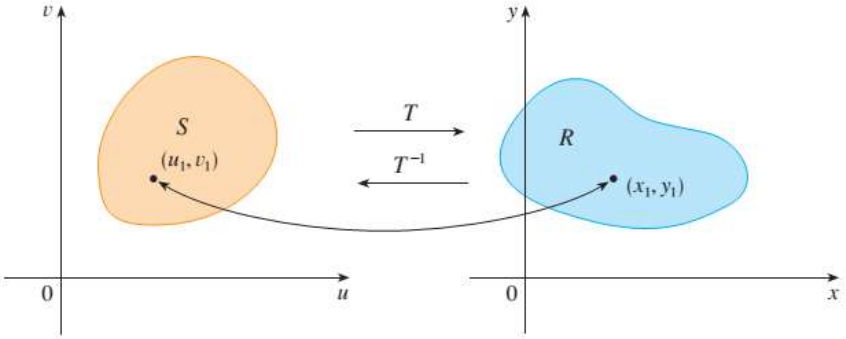


FIGURE 1

If T is a one-to-one transformation, then it has an **inverse transformation** T^{-1} from the xy-plane to the uv-plane and it may be possible to solve Equations 3 for u and v in terms of x and y:

$$u = G(x, y) \qquad v = H(x, y)$$

EXAMPLE 1 A transformation is defined by the equations

$$x = u^2 - v^2 \qquad y = 2uv$$

Find the image of the square $S = \{(u, v) \mid 0 \leq u \leq 1,\ 0 \leq v \leq 1\}$.

SOLUTION The transformation maps the boundary of S into the boundary of the image. So we begin by finding the images of the sides of S. The first side, S_1, is given by $v = 0$ $(0 \leq u \leq 1)$. (See Figure 2.) From the given equations we have $x = u^2$, $y = 0$, and so $0 \leq x \leq 1$. Thus S_1 is mapped into the line segment from (0, 0) to (1, 0) in the xy-plane. The second side, S_2, is $u = 1$ $(0 \leq v \leq 1)$ and, putting $u = 1$ in the given equations, we get

$$x = 1 - v^2 \qquad y = 2v$$

Eliminating v, we obtain

$$x = 1 - \frac{y^2}{4} \qquad 0 \leq x \leq 1 \tag{4}$$

which is part of a parabola. Similarly, S_3 is given by $v = 1$ $(0 \leq u \leq 1)$, whose image is the parabolic arc

$$x = \frac{y^2}{4} - 1 \qquad -1 \leq x \leq 0 \tag{5}$$

Finally, S_4 is given by $u = 0$ $(0 \leq v \leq 1)$ whose image is $x = -v^2$, $y = 0$, that is, $-1 \leq x \leq 0$. (Notice that as we move around the square in the counterclockwise direction, we also move around the parabolic region in the counterclockwise direction.) The image of S is the region R (shown in Figure 2) bounded by the x-axis and the parabolas given by Equations 4 and 5. ■

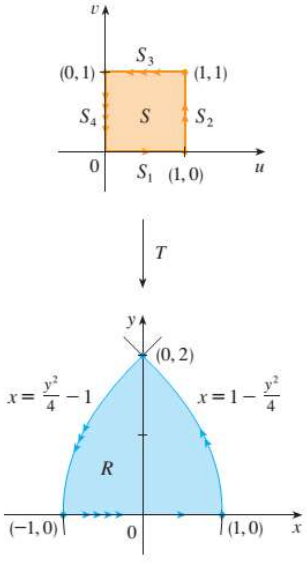


FIGURE 2

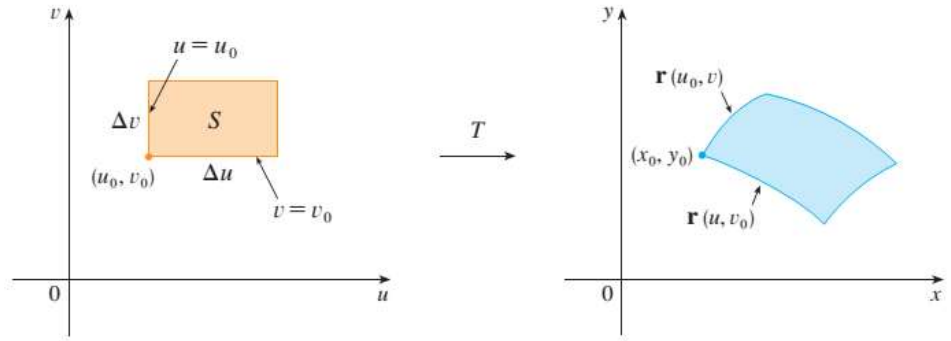


FIGURE 3

The image of S is a region R in the xy-plane, one of whose boundary points is $(x_0, y_0) = T(u_0, v_0)$. The vector

$$\mathbf{r}(u, v) = g(u, v)\,\mathbf{i} + h(u, v)\,\mathbf{j}$$

is the position vector of the image of the point (u, v). The equation of the lower side of S is $v = v_0$, whose image curve is given by the vector function $\mathbf{r}(u, v_0)$. The tangent vector at (x_0, y_0) to this image curve is

$$\mathbf{r}_u = g_u(u_0, v_0)\,\mathbf{i} + h_u(u_0, v_0)\,\mathbf{j} = \frac{\partial x}{\partial u}\,\mathbf{i} + \frac{\partial y}{\partial u}\,\mathbf{j}$$

Similarly, the tangent vector at (x_0, y_0) to the image curve of the left side of S (namely, $u = u_0$) is

$$\mathbf{r}_v = g_v(u_0, v_0)\,\mathbf{i} + h_v(u_0, v_0)\,\mathbf{j} = \frac{\partial x}{\partial v}\,\mathbf{i} + \frac{\partial y}{\partial v}\,\mathbf{j}$$

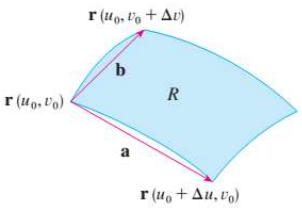


FIGURE 4

We can approximate the image region $R = T(S)$ by a parallelogram determined by the secant vectors

$$\mathbf{a} = \mathbf{r}(u_0 + \Delta u, v_0) - \mathbf{r}(u_0, v_0) \qquad \mathbf{b} = \mathbf{r}(u_0, v_0 + \Delta v) - \mathbf{r}(u_0, v_0)$$

shown in Figure 4. But

$$\mathbf{r}_u = \lim_{\Delta u \to 0} \frac{\mathbf{r}(u_0 + \Delta u, v_0) - \mathbf{r}(u_0, v_0)}{\Delta u}$$

and so $$\mathbf{r}(u_0 + \Delta u, v_0) - \mathbf{r}(u_0, v_0) \approx \Delta u\,\mathbf{r}_u$$

Similarly $$\mathbf{r}(u_0, v_0 + \Delta v) - \mathbf{r}(u_0, v_0) \approx \Delta v\,\mathbf{r}_v$$

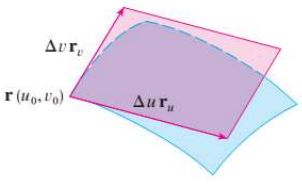


FIGURE 5

This means that we can approximate R by a parallelogram determined by the vectors $\Delta u\,\mathbf{r}_u$ and $\Delta v\,\mathbf{r}_v$. (See Figure 5.) Therefore we can approximate the area of R by the area of this parallelogram, which, from Section 12.4, is

7 DEFINITION The **Jacobian** of the transformation T given by $x = g(u, v)$ and $y = h(u, v)$ is

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \dfrac{\partial x}{\partial u} & \dfrac{\partial x}{\partial v} \\ \dfrac{\partial y}{\partial u} & \dfrac{\partial y}{\partial v} \end{vmatrix} = \frac{\partial x}{\partial u}\frac{\partial y}{\partial v} - \frac{\partial x}{\partial v}\frac{\partial y}{\partial u}$$

With this notation we can use Equation 6 to give an approximation to the area ΔA of R:

$$\Delta A \approx \left|\frac{\partial(x, y)}{\partial(u, v)}\right| \Delta u\, \Delta v \tag{8}$$

where the Jacobian is evaluated at (u_0, v_0).

Next we divide a region S in the uv-plane into rectangles S_{ij} and call their images in the xy-plane R_{ij}. (See Figure 6.)

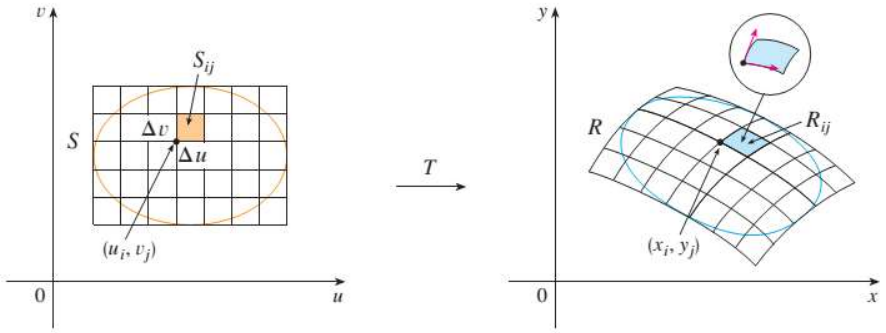


FIGURE 6

Applying the approximation (8) to each R_{ij}, we approximate the double integral of f over R as follows:

$$\begin{aligned} \iint_R f(x, y)\, dA &\approx \sum_{i=1}^{m}\sum_{j=1}^{n} f(x_i, y_j)\, \Delta A \\ &\approx \sum_{i=1}^{m}\sum_{j=1}^{n} f\big(g(u_i, v_j), h(u_i, v_j)\big)\left|\frac{\partial(x, y)}{\partial(u, v)}\right| \Delta u\, \Delta v \end{aligned}$$

where the Jacobian is evaluated at (u_i, v_j). Notice that this double sum is a Riemann sum for the integral

$$\iint_S f\big(g(u, v), h(u, v)\big)\left|\frac{\partial(x, y)}{\partial(u, v)}\right| du\, dv$$

The foregoing argument suggests that the following theorem is true. (A full proof is given in books on advanced calculus.)

9 CHANGE OF VARIABLES IN A DOUBLE INTEGRAL Suppose that T is a C^1 transformation whose Jacobian is nonzero and that maps a region S in the uv-plane onto a region R in the xy-plane. Suppose that f is continuous on R and that R and S are type I or type II plane regions. Suppose also that T is one-to-one, except perhaps on the boundary of S. Then

$$\iint_R f(x, y)\, dA = \iint_S f\big(x(u, v), y(u, v)\big) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du\, dv$$

Theorem 9 says that we change from an integral in x and y to an integral in u and v by expressing x and y in terms of u and v and writing

$$dA = \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du\, dv$$

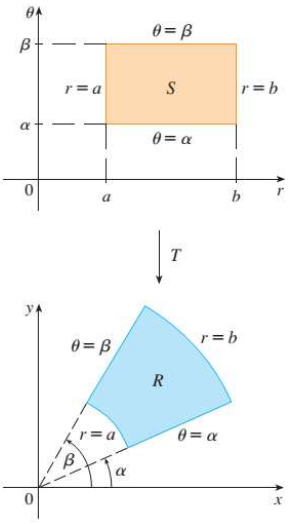


FIGURE 7
The polar coordinate transformation

As a first illustration of Theorem 9, we show that the formula for integration in polar coordinates is just a special case. Here the transformation T from the $r\theta$-plane to the xy-plane is given by

$$x = g(r, \theta) = r\cos\theta \qquad y = h(r, \theta) = r\sin\theta$$

and the geometry of the transformation is shown in Figure 7. T maps an ordinary rectangle in the $r\theta$-plane to a polar rectangle in the xy-plane. The Jacobian of T is

$$\frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \dfrac{\partial x}{\partial r} & \dfrac{\partial x}{\partial \theta} \\ \dfrac{\partial y}{\partial r} & \dfrac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r\cos^2\theta + r\sin^2\theta = r > 0$$

Thus Theorem 9 gives

$$\begin{aligned} \iint_R f(x, y)\, dx\, dy &= \iint_S f(r\cos\theta, r\sin\theta) \left| \frac{\partial(x, y)}{\partial(r, \theta)} \right| dr\, d\theta \\ &= \int_\alpha^\beta \int_a^b f(r\cos\theta, r\sin\theta)\, r\, dr\, d\theta \end{aligned}$$

which is the same as Formula 15.4.2.

EXAMPLE 2 Use the change of variables $x = u^2 - v^2$, $y = 2uv$ to evaluate the integral $\iint_R y\,dA$, where R is the region bounded by the x-axis and the parabolas $y^2 = 4 - 4x$ and $y^2 = 4 + 4x$, $y \geqslant 0$.

SOLUTION The region R is pictured in Figure 2 (on page 1014). In Example 1 we discovered that $T(S) = R$, where S is the square $[0, 1] \times [0, 1]$. Indeed, the reason for making the change of variables to evaluate the integral is that S is a much simpler region than R. First we need to compute the Jacobian:

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \dfrac{\partial x}{\partial u} & \dfrac{\partial x}{\partial v} \\ \dfrac{\partial y}{\partial u} & \dfrac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} 2u & -2v \\ 2v & 2u \end{vmatrix} = 4u^2 + 4v^2 > 0$$

Therefore, by Theorem 9,

$$\begin{aligned} \iint_R y\,dA &= \iint_S 2uv \left|\frac{\partial(x, y)}{\partial(u, v)}\right| dA = \int_0^1 \int_0^1 (2uv)4(u^2 + v^2)\,du\,dv \\ &= 8\int_0^1 \int_0^1 (u^3v + uv^3)\,du\,dv = 8\int_0^1 \left[\tfrac{1}{4}u^4v + \tfrac{1}{2}u^2v^3\right]_{u=0}^{u=1} dv \\ &= \int_0^1 (2v + 4v^3)\,dv = \left[v^2 + v^4\right]_0^1 = 2 \end{aligned}$$

EXAMPLE 3 Evaluate the integral $\iint_R e^{(x+y)/(x-y)}\,dA$, where R is the trapezoidal region with vertices $(1, 0)$, $(2, 0)$, $(0, -2)$, and $(0, -1)$.

SOLUTION Since it isn't easy to integrate $e^{(x+y)/(x-y)}$, we make a change of variables suggested by the form of this function:

$$u = x + y \qquad v = x - y \tag{10}$$

These equations define a transformation T^{-1} from the xy-plane to the uv-plane. Theorem 9 talks about a transformation T from the uv-plane to the xy-plane. It is obtained by solving Equations 10 for x and y:

$$x = \tfrac{1}{2}(u + v) \qquad y = \tfrac{1}{2}(u - v) \tag{11}$$

The Jacobian of T is

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \dfrac{\partial x}{\partial u} & \dfrac{\partial x}{\partial v} \\ \dfrac{\partial y}{\partial u} & \dfrac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{vmatrix} = -\tfrac{1}{2}$$

To find the region S in the uv-plane corresponding to R, we note that the sides of R lie on the lines

$$y = 0 \qquad x - y = 2 \qquad x = 0 \qquad x - y = 1$$

and, from either Equations 10 or Equations 11, the image lines in the uv-plane are

$$u = v \qquad v = 2 \qquad u = -v \qquad v = 1$$

Thus the region S is the trapezoidal region with vertices $(1, 1)$, $(2, 2)$, $(-2, 2)$, and $(-1, 1)$ shown in Figure 8. Since

$$S = \{(u, v) \mid 1 \leq v \leq 2,\ -v \leq u \leq v\}$$

Theorem 9 gives

$$\begin{aligned} \iint_R e^{(x+y)/(x-y)}\,dA &= \iint_S e^{u/v}\left|\frac{\partial(x, y)}{\partial(u, v)}\right| du\,dv \\ &= \int_1^2 \int_{-v}^{v} e^{u/v}\left(\tfrac{1}{2}\right) du\,dv = \tfrac{1}{2}\int_1^2 \left[ve^{u/v}\right]_{u=-v}^{u=v} dv \\ &= \tfrac{1}{2}\int_1^2 (e - e^{-1})v\,dv = \tfrac{3}{4}(e - e^{-1}) \end{aligned}$$

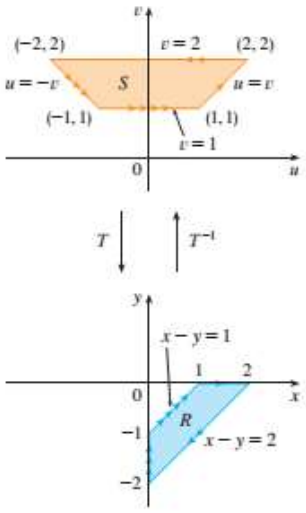


FIGURE 8

TRIPLE INTEGRALS

There is a similar change of variables formula for triple integrals. Let T be a transformation that maps a region S in uvw-space onto a region R in xyz-space by means of the equations

$$x = g(u, v, w) \qquad y = h(u, v, w) \qquad z = k(u, v, w)$$

The **Jacobian** of T is the following 3×3 determinant:

12
$$\frac{\partial(x, y, z)}{\partial(u, v, w)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix}$$

Under hypotheses similar to those in Theorem 9, we have the following formula for triple integrals:

13
$$\iiint_R f(x, y, z)\, dV = \iiint_S f\big(x(u, v, w), y(u, v, w), z(u, v, w)\big) \left| \frac{\partial(x, y, z)}{\partial(u, v, w)} \right| du\, dv\, dw$$

6.) Let R be the region in xyz-space defined by the inequalities $1 \le x \le 2$, $0 \le xy \le 2$, $0 \le z \le 1$. Evaluate the integral below by first applying the transformation $u = x$, $v = xy$, $w = 3z$.

$$\int\int\int_R (x^2 y + 3xyz) dx dy dz\,.$$

Solution:

The inverse transformation is $x = u$, $y = v/u$ and $z = w/3$. The region is mapped into $1 \le u \le 2$, $0 \le v \le 2$, $0 \le w \le 3$, and calculating the Jacobian we find $J = 1/(3u)$ which is positive. Therefore the integral becomes,

$$\begin{aligned} I &= \int_1^2 \int_0^3 \int_0^2 (uv + vw) \frac{1}{3u} dv dw du \\ &= \int_1^2 \int_0^3 (2u + 2w) \frac{1}{3u} dw du \\ &= \int_1^2 (6u + 9) \frac{1}{3u} du \\ &= 2 + 3 \ln 2 \end{aligned}$$

3. Evaluate the following integrals:

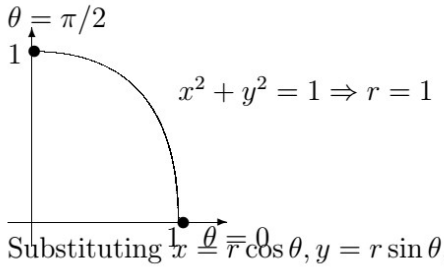
(1) $\displaystyle\int_0^1 \int_0^{\sqrt{1-x^2}} (x^2+y^2)dy\,dx$

(2) $\displaystyle\iint_R e^{x^2+y^2}\,dy\,dx$

where R is the semicircular region bounded by the x-axis and the curve $y=\sqrt{1-x^2}$.

Solution:

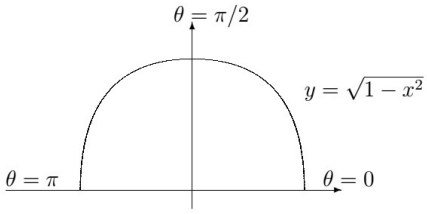
(1) It is easier to evaluate this double integral in polar coordinates. In order to determine limits of integration let's sketch the region of integration.



Substituting $x=r\cos\theta, y=r\sin\theta$, and replacing $dx\,dy$ by $r\,dr\,d\theta$, we get

$$\begin{aligned}\int_0^1 \int_0^{\sqrt{1-x^2}} (x^2+y^2)dy\,dx &= \int_0^{\pi/2}\int_0^1 r^2\, r\,dr\,d\theta \\ &= \int_0^{\pi/2}\left[\frac{r^4}{4}\right]_0^1 d\theta \\ &= \int_0^{\pi/2}\frac{1}{4}\,d\theta = \frac{\pi}{8}\end{aligned}$$

(2) In Cartesian coordinates, the integral in question is a non-elementary integral and there is no direct way to integrate $e^{x^2+y^2}$ with respect to x or y. Again we will apply polar coordinates.



The region R is substituting $x=r\cos\theta, y=r\sin\theta$ and replacing $dy\,dx$ by $r\,dr\,d\theta$ enables us to evaluate the integral as

$$\begin{aligned}\iint_R e^{x^2+y^2}dy\,dx &= \int_0^{\pi}\int_0^1 e^{r^2} r\,dr\,d\theta \\ &= \int_0^{pi}\left[\frac{e^{r^2}}{2}\right]_0^1 d\theta \\ &= \int_0^{\pi}\frac{1}{2}(e-1)d\theta = \frac{\pi}{2}(e-1)\end{aligned}$$

4. Evaluate

$$\int_0^3 \int_0^4 \int_{y/2}^{(y/2)+1} \left(\frac{2x-y}{2} + \frac{z}{3} \right) dx\, dy\, dz$$

by applying the transformation $u = \dfrac{(2x-y)}{2},\ v = \dfrac{y}{2},\ w = \dfrac{z}{3}$

Solution:

Let's call the region of integration in xyz-space as D and corresponding uvw-region as G. We need to find the boundaries of the region in uvw-region and the Jacobian of the transformation in order to evaluate this integral. To find them, we first solve equations $u = \dfrac{(2x-y)}{2},\ v = \dfrac{y}{2},\ w = \dfrac{z}{3}$ for x,y and z in terms of u,v and w. Routine algebra gives:

$$\begin{aligned} x &= u+v \\ y &= 2v \\ z &= 3w \end{aligned}$$

We then find the boundaries of G by substituting these expressions into the equations for the boundaries of D.

xyz-equations for the boundary of D	Corresponding uvw-equations for the boundary of G	Simplified uvw-equations
$x = \dfrac{y}{2}$	$u+v = \dfrac{2v}{2} = v$	$u = 0$
$x = \dfrac{y}{2} + 1$	$u+v = \left(\dfrac{2v}{2}\right) + 1 = v+1$	$u = 1$
$y = 0$	$2v = 0$	$v = 0$
$y = 4$	$2v = 4$	$v = 2$
$z = 0$	$3w = 0$	$w = 0$
$z = 3$	$3w = 3$	$w = 1$

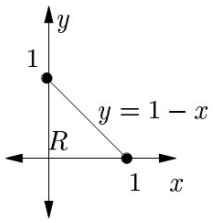
The Jacobian of the transformation is

$$\hat{J}(u,v,w) = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix} = \begin{vmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{vmatrix} = 6$$

$$\begin{aligned} \int_0^3 \int_0^4 \int_{y/2}^{(y/2)+1} \left(\frac{2x-y}{2} + \frac{z}{3}\right) dx\, dy\, dz &= \int_0^1 \int_0^2 \int_0^1 (u+w)|\hat{J}(u,v,w)| du\, dv\, dw \\ &= \int_0^1 \int_0^2 \int_0^1 (u+w)6\, du\, dv\, dw \\ &= 6\int_0^1 \int_0^2 \left[\frac{u^2}{2} + uw\right]_0^1 dv\, dw \\ &= 6\int_0^1 \int_0^2 \left(\frac{1}{2} + w\right) dv\, dw \\ &= 6\int_0^1 \left(\frac{v}{2} + vw\right)_0^2 dw = 6\int_0^1 (1+2w) dw \\ &= 6\Big[w + w^2\Big]_0^1 = 6 \cdot 2 = 12 \end{aligned}$$

5.) [20] Find the volume of the solid bounded by the paraboloid $z = x^2 + y^2 + 4$ and the planes $x = 0$, $y = 0$, $z = 0$, $x + y = 1$.

Solution:

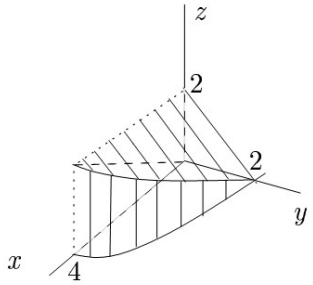


$$
\begin{aligned}
V &= \int_0^1 \int_0^{1-x} (x^2 + y^2 + 4) dy dx \\
&= \int_0^1 \left[x^2 y + \frac{y^3}{3} + 4y \right]_0^{1-x} dx \\
&= \int_0^1 \left[x^2(1-x) + \frac{(1-x)^3}{3} + 4(1-x) \right] dx \\
&= \int_0^1 \left[x^2 - x^3 + \frac{1}{3}(1 - 3x + 3x^2 - x^3) + 4 - 4x \right] dx \\
&= \int_0^1 \left(-\frac{4}{3}x^3 + 2x^2 - 5x + \frac{13}{3} \right) dx \\
&= \left[-\frac{4}{3}\frac{x^4}{4} + \frac{2x^3}{3} - \frac{5x^2}{2} + \frac{13x}{3} \right]_0^1 \\
&= -\frac{1}{3} + \frac{2}{3} - \frac{5}{2} + \frac{13}{3} \\
&= \frac{14}{3} - \frac{5}{2} \\
&= \frac{13}{6}
\end{aligned}
$$

4. Let R be the solid region in the first octant bounded by the coordinate planes, the plane $y + z = 2$ and the cylinder $x = 4 - y^2$. Express the volume of R as an iterated triple integral:

(a) in $dzdxdy$ order (do not evaluate).

Solution:



The solid is bounded by the coordinate planes which are $x = 0, y = 0, z = 0$. Hence easily:

$$V(R) = \int_0^2 \int_0^{4-y^2} \int_0^{2-y} dzdxdy.$$

(b) in $dydxdz$ order (do not evaluate).

Solution:

Along the y-axis (slices parallel to xz-plane) there are two possible boundaries. So this is the sum two triple integrals:

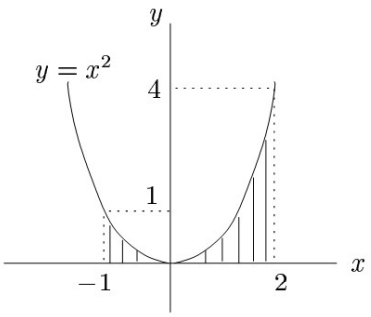
$$V(R) = \iint_{D_1} \int_0^{2-z} dydA + \iint_{D_2} \int_0^{\sqrt{4-x}} dydA,$$

where $D_1 = \{(x, z)|0 \leq x \leq 4 - (2 - z)^2,\ 0 \leq z \leq 2\}$ and $D_2 = \{(x, z)|4 - (2 - z)^2 \leq x \leq 4,\ 0 \leq z \leq 2\}$. Hence:

$$V(R) = \int_0^2 \int_0^{4-(2-z)^2} \int_0^{2-z} dydxdz + \int_0^2 \int_{4-(2-z)^2}^4 \int_0^{\sqrt{4-x}} dydxdz.$$

1. Evaluate $\int_0^1 \int_{-1}^{-\sqrt{y}} \cos 3x^3 dxdy + \int_0^4 \int_{\sqrt{y}}^2 \cos 3x^3 dxdy.$

Solution:



Changing the order of iteration over the shaded region in the above figure, one obtains:

$$\begin{aligned}\left(\int_0^1 \int_{-1}^{-\sqrt{y}} + \int_0^4 \int_{\sqrt{y}}^2\right) \cos 3x^3 dxdy &= \int_{-1}^2 \int_0^{x^2} \cos 3x^3 dydx \\ &= \int_{-1}^2 x^2 \cos 3x^3 dx \\ &= \frac{1}{9} \sin 3x^3 \Big|_{-1}^2 \\ &= \frac{1}{9} \left(\sin(24) - \sin(-3)\right).\end{aligned}$$

3. Evaluate:
(a)

$$\int_0^1 \int_y^1 \frac{\sin x}{x} dxdy.$$

(b)

$$\int_{-1}^1 \int_{-\sqrt{1-y^2}}^0 \frac{4\sqrt{x^2+y^2}}{1+x^2+y^2} dxdy.$$

Solution:

(a)
$$\begin{aligned}\int_0^1 \int_y^1 \frac{\sin x}{x} dxdy &= \int_0^1 \int_0^x \frac{\sin x}{x} dydx = \int_0^1 \frac{\sin x}{x} y \Big|_0^x dx \\ &= \int_0^1 \sin x dx = \cos x|_0^1 = -\cos 1 + 1.\end{aligned}$$

(b) Let us use polar coordinates. So, by putting $x^2 + y^2 = r^2$,

$$\begin{aligned}\int_{-1}^1 \int_{-\sqrt{1-y^2}}^0 \frac{4\sqrt{x^2+y^2}}{1+x^2+y^2} dxdy &= \int_{\pi/2}^{3\pi/2} \int_0^1 \frac{4r}{1+r^2} r dr d\theta \\ &= (3\pi/2 - \pi/2) \int_0^1 \frac{4r^2+4-4}{1+r^2} dr d\theta \\ &= \pi \int_0^1 \left(4 - 4\frac{1}{1+r^2}\right) dr d\theta \\ &= \pi \left(4r - 4\tan^{-1} r\right)_0^1 = \pi(4-\pi) \\ &= 4\pi - \pi^2.\end{aligned}$$